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Power Electronics Coursework

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Report

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Introduction

The purpose of this report is to outline the choices made throughout the initial design stage of the boost converter for the power electronics coursework. This reports is structured around the principle of iterative design. We begin by analysing the specifications and designing an ideal circuit, such that the specification is achieved. We use the collected data, observations and simulations to help us in the development of a real circuit with real components for the open-loop boost converter stage. We consider the non-idealities which are introduced as a consequence, and make appropriate modifications to the design in order to insure that the initial specifications have been met. Pspice simulations are used to test our analytic predictions and confirm that the design is satisfactory. Different methods of performance improvement, in particular efficiency, are considered, such the replacement of the power diode by a second MOSFET, in order to minimise dissipation losses and undesired voltage drops. Finally, we consider the introduction of feedback into the system, and design a voltage-mode compensator to ensure system stability for all frequencies and input voltage variations, as well as improved system performance; in particular improvements in output voltage settling times and the constancy of output voltage.

Design Specifications Analysis

This section outlines our understanding and consideration of the design specifications, and analyses their possible implications and impact on the design.

- *1 Input voltage: 20 V (±2 V).*

We must be aware of the variations in input voltage during steady – state operation. This implies that there may exist a voltage ripple on the input voltage supply throughout steady-state operation, as well as there may be variations in the dc level of input voltage between 18 Volts and 22 Volts.

- *2 Output voltage: 48 V (±0.5 V ripple).*

The output voltage dc level is said to remain at 48 V throughout circuit operation, however the specification allows for an ac voltage ripple of 1V in amplitude at the output terminal, i.e. output voltage may oscillate between 47.5 and 48.5 V.

- *3 Maximum output current: 0.6 A, continuous, resistive loads only.*

The maximum load current which should be supported by the design is 0.6 A. The fact that the load is resistive implies a power factor of 1, and that all of the energy supplied by the boost converter is consumed by the load. Therefore the average power output should be rated at 48V x 0.6A = 28.8 A.

- *4 Input Current ripple: max 10% at maximum output power.*

Output current is related to the input current through the duty cycle, and input current increases together with output current. Since the dc output voltage is constant, then maximum power output must occur at maximum permissible output current. In this case, the amplitude of the ripple current should be at most 10% of the average input (dc) current.

- *5 A Soft-Start to limit inrush current.*

As can be seen in the later simulations, abnormally high input currents appear during the transitions of the converter from the off-state into the on-state. These dc current overshoots can be as high as 15 times the maximum permissible input currents (depends on the particular circuit configuration) and could easily exceed the component ratings and damage the device. We must thus ensure that the transition into steady state is limited and that there exists a controllable limit on the maximum allowable inrush current.

- *6 Switching frequency: <200 kHz*

The choice of switching frequency completely determines the minimum permissible component sizes for our design, such that the specification is achieved. We must thus select the frequency based upon the real components that will be available to us. Higher frequencies will improve the transient response, but imply faster device switching and thus lower efficiencies due to higher switching losses, as well as the introduction additional switching noise into our system. Frequency is limited by the MOSFET switching times, as well as the gate power drivers, the delay times of which are typically greater than the MOSFET switching times.

• 7 Efficiency > 90% at full load

Our boost converter must be efficient and unnecessary power losses must be avoided. Design choices should be made with a bias towards low power dissipation and low voltage drops. This specification will be particularly relevant when selecting the semiconductors, prioritising lower voltage drops for diodes and lower drain-source resistances for the MOSFET.

• 8 Maximum inductor volume of 9 cm3

The size of the inductor completely determines the amplitude of the input current ripple once the frequency of operation has been chosen and the input voltage defined. Thus our inductor must be large enough so that we achieve the input current ripple requirement. However, the maximum core size of the inductor has been specified, so we will have to look towards other means of reducing input current oscillations, in particular increasing the operating frequency (but not higher than the permissible frequency).

2 Basic Converter Design

Here we design the circuit assuming that the components we are using are ideal. We thus discard parameters such as parasitics or component ratings.

Arguably the central component of the boost converter is the inductor, since it defines the size of the current ripple, and also affects output voltage ripple. It should thus be acceptable to start our design by picking the inductor first. We thus attempt to estimate the minimum inductor size which we may use.

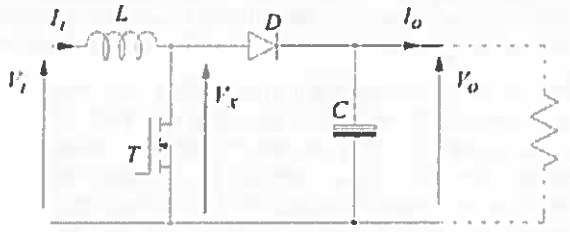


Figure 1

During steady-state operation of a boost converter we must consider both the continuous and discontinuous modes. Requirement 4 places a threshold on the maximum possible ripple current. In order to calculate the minimum inductance value, we must choose the worst possible limiting case for the minimum inductor size, which is the maximum allowable current ripple. Maximum input current ripple will occur in continuous mode, since in discontinuous mode the inductor current remains zero for some time throughout each switching cycle, naturally making the ripple amplitude less.

The maximum allowable ripple is defined as 10% of input current at maximum power output. This occurs when the output current, I_o, equals 0.6A. So, in order to calculate what the ripple would be in this case, we must calculate the input current as a function of output current. The two dc currents are related to each other through the following expression, which is a parameter of duty cycle.

$$I_{IN} = \frac{1}{1-\delta} I_o$$

Hence the inductor ripple current must satisfy:

$$\Delta I < 0.1 I_{INmax} = \frac{0.1}{1-\delta} I_{Omax}$$

But since $\Delta I = \frac{V_I}{L} t_{on}$, where ton is the "MOSFET on" time:

$$L > \frac{10(1-\delta)\delta V_{IN}}{f I_{Omax}}$$

We must now maximise this expression in order to calculate the minimum allowable inductance value. To do this we maximise Vin and must also maximise$(1-\delta)\delta$. This function has a maximum point at δ = 0.5. However, we will operate at a range of duty cycles, which is higher than 0.5. We must find the range of duty cycles we will use in order to correctly maximise the expression for minimum inductance. Using the textbook principle of adding the "on current slope" and "off current slope" to get 0, we acquire the duty cycle expression. The diode voltage drop has been retained in the expression deliberately, since it introduces a large

inaccuracy in the duty cycle estimation for a particular output voltage. For a power diode, the forward drop is commonly approximated to be 1V.

$$\delta = \frac{V_O + V_{diode} - V_I}{V_O + V_{diode}}$$

We may now calculate the minimum and maximum duty cycles we may have to use in our circuit.

$$0.551 = \frac{48 + 1 - 22}{48 + 1} = \frac{V_O + V_{diode} - V_{INmax}}{V_O + V_{diode}} < \delta < \frac{V_O + V_{diode} - V_{INmin}}{V_O + V_{diode}} = \frac{48 + 1 - 18}{48 + 1} = 0.633$$

0.551 is closest to 0.5, and will thus maximise$(1 - \delta)\delta$. Hence the minimum inductor value:

$$L = \frac{10(1 - 0.551)551 * 22}{f * 0.6}$$

We must now specify the frequency of operation of our converter. Since we may choose f to be anywhere between 0 and 200kHz, a good guess would be halfway at 100kHZ. Such a value of frequency will be high enough to yield a reasonably-sized inductor which could be purchased in stores, and low enough for the MOSFET to be able to switch at that frequency with bearable switching loses. Thus $L = 907\mu H$.

Please also note that in a realistic scenario the converter will be less efficient. The higher the inefficiency, the higher is the maximum input current required to supply the maximum load, hence the larger the allowable ripple for the input current for less efficient converters, which leads to a smaller value of minimum inductance. Thus we would typically be able to do with an even smaller inductor and still achieve the specification.

Now that we have chosen the inductor value, we may proceed to choose the minimum capacitance value for the output capacitor. The capacitor must be large enough to keep the output voltage as constant as possible; that is constant enough that specification 2 is achieved. The output voltage must be 48V with up to 1V amplitude voltage ripple. If we assume that only the dc current (Io) flows into the load, then all of the ac ripple must flow through the capacitor C.

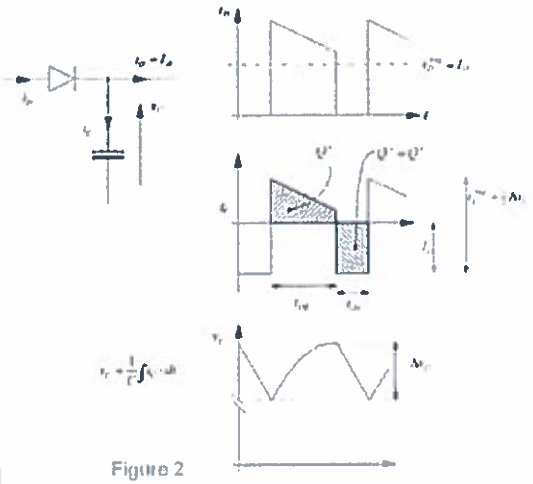


Figure 2

By using the standard capacitor differential equation, we realise that the voltage ripple across it is simply given by $v_c = \frac{1}{C}\int i_c\, dt$, and thus the amplitude of the voltage ripple is $\Delta v_c = \frac{1}{C}Q^+ = \frac{1}{C}Q^- = \frac{1}{C}I_O t_{on} = \frac{I_O \delta}{fC} < 1$

Solving for maximum C: $C > \frac{I_{Omax}\delta max}{f} = \frac{0.6 \cdot 0.633}{100k} = 3.80\mu F$. This is a very small capacitor for an SMPS. It is so small due to the fact that Io is relatively small for an SMPS, while f is high. We will later see why such a small capacitor is unreasonable given realistic components.

Now that we have selected the inductor and capacitor values, we may proceed to calculate the ideal component ratings for the inductor, diode and MOSFET, assuming that the converter is operating in steady-state. We will account for the transient boost converter response when we select the real components later on.

Inductor ratings

Real inductors have two very important parameters, which are relevant to our design: RMS inductor current and inductor saturation current. Irms is normally specified as the dc current that produces an inductor temperature rise of 40°C. Effectively it means that we should not apply a higher dc current to an inductor than its rated rms current, since the excess heat may damage the component or change its properties. Saturation current is relevant to non-air core inductors, such as the one we will use. At currents above the saturation current the core material can no longer linearly increase the amount of magnetic flux with an incremental

increase in current. This results in the effective inductance dropping and causes further increases in current, making the inductor behave closer to an open circuit. We should never have the inductor current exceed the saturation current, since this will lead to inefficiencies and altered circuit behaviour.

The maximum peak inductor current should be smaller than the inductor saturation current. The maximum peak current must occur in continuous mode for our circuit. Assuming steady state and maximum load current of 0.6A it would be given by the sum of the average dc current and half of the current ripple. I.e. for a duty cycle of 0.551 (the δ we would use for Vinmax)

$$I_{Lpeak} = I_{INmax} + \frac{1}{2}\frac{V_{INmax}\delta}{Lf} = \frac{I_{Omax}}{1-\delta} + \frac{1}{2}\frac{V_{INmax}\delta}{Lf} = 1.40A$$

The maximum RMS inductor current is simply equal to the maximum dc inductor current, which is given by

$$I_{RMSmax} = I_{INmax} = \frac{I_{Omax}}{1-\delta} = \frac{0.6}{1-0.551} = 1.34A$$

And should be less than the RMS rated current for the inductor.

MOSFET ratings

When the MOSFET is in its off state, it must be able to tolerate a drain-source voltage of $V_O + V_{diode}$ (check Figure 1). This would imply approximately 48V + 1V (diode forward drop) +0.5V (maximum allowed voltage drop by requirement 2) = 49.5V. This quantity is only relevant for the steady state.

The MOSFET must be able to tolerate drain-source currents equal to the peak inductor current. This is because during the on-state the same current flows through the MOSFET as through the inductor. Hence the MOSFET must be able to handle a current of 1.4A.

Diode rating

When the MOSFET is on, the diode will be reverse-biased. It will therefore have to block a voltage which is approximately equal to the output voltage minus the drain-source MOSFET voltage drop. Since the drain-source FET resistance can be considered small, the approximate blocking voltage should be up to 48V + 0.5V(ripple) = 49V. The peak diode current should be the same as the peak inductor current, i.e. 1.40A.

3 Simulation

Efficiency

In this section we will simulate the ideal (not really) boost converter using Orcad Pspice. The components C, L and R form the "Analog" library will be used for the capacitor, inductor and resistor respectively. Dbreak will be used to represent the power diode and the switch S will be used instead of the MOSFET.

Please note that since this is an ideal component simulation, the switch on-resistance has been set to 0 instead of its predefined value. Also note that the inductor and capacitor still dissipate a very small amount of power according to their model parameters. The diode has a forward voltage drop of approximately 0.9V for high currents, thus providing the largest source of inefficiency in the "ideal circuit". We will see that the measured efficiency of the circuit is not 100%.

Firstly we will measure the efficiency of the boost converter under a range of operating conditions. This can be done by measuring the average power dissipated by the input voltage source and the output load using the power meter in steady-state. The output power can then be compared to the input power. Figure 3 shows the schematic which has been used to measure the efficiency performance of the "ideal" boost circuit with the following parameters:

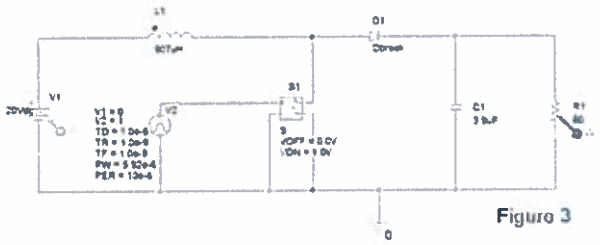


Figure 3

Table 1

L1	907uH
C1	3.8uF
R1	80Ohms
V1	20Vdc
Duty cycle	0.592
Frequency	100kHz

These parameters have been selected to test the boost at max load with typical input voltage.

Before testing efficiency we must make sure the model behaves as predicted. With the parameters set up as above the inductor current and load currents are shown on Figure 4. We immediately see that the average inductor current is 1.34A with 10% ripple at an output current of 600mA. If we had not included the diode forward drop in our calculations for the required duty cycle, then the estimations would not have been so accurate. We will now test efficiency for a range of parameters.

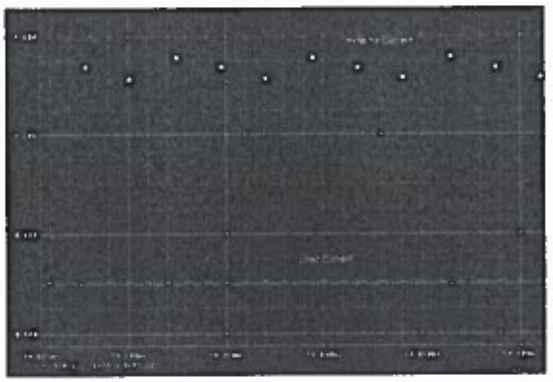


Figure 4

Table 2

	efficiency
R1=80Ohms→Io=0.6A	η= 0.979
R1=96Ohms→Io=0.5A	η= 0.980
R1=120Ohms→Io=0.4A	η= 0.981
R1=160Ohms→Io=0.3A	η= 0.981
R1=240Ohms→Io=0.2A	η= 0.982
R1=480Ohms→Io=0.1A	η= 0.983

We note that efficiency is not 100% as we may have expected. This is mostly due to the non-zero diode forward voltage drop. We also note that efficiency slightly decreases with increasing current. Most likely cause is higher powers being dissipated within the diode, since the diode dissipation power is the product of its forward drop and the current flowing through it. Note that if the diode dissipation power is simulated, it will almost entirely account for the loss in efficiency.

Start-up simulation

To test the start-up transient behaviour we set the inductor and capacitor initial conditions to zero. The circuit parameters are the same as in table 1 (i.e. maximum load). Figure 5 shows the output voltage transient and Figure 6 shows the inductor current transient.

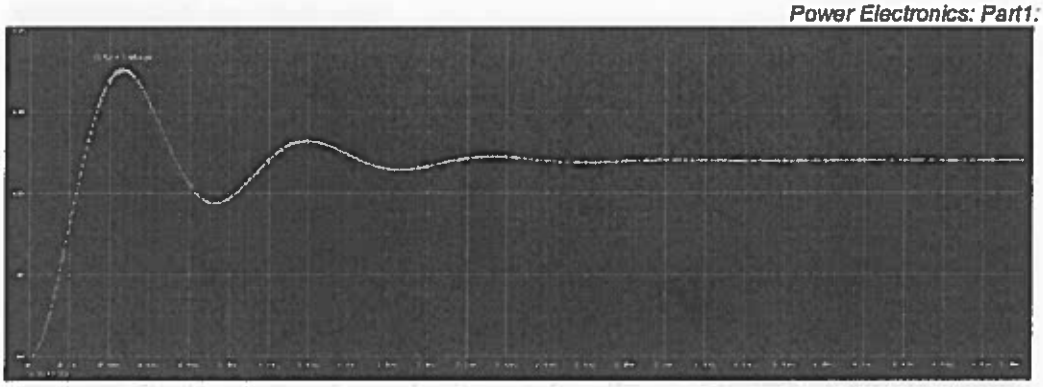


Figure 5

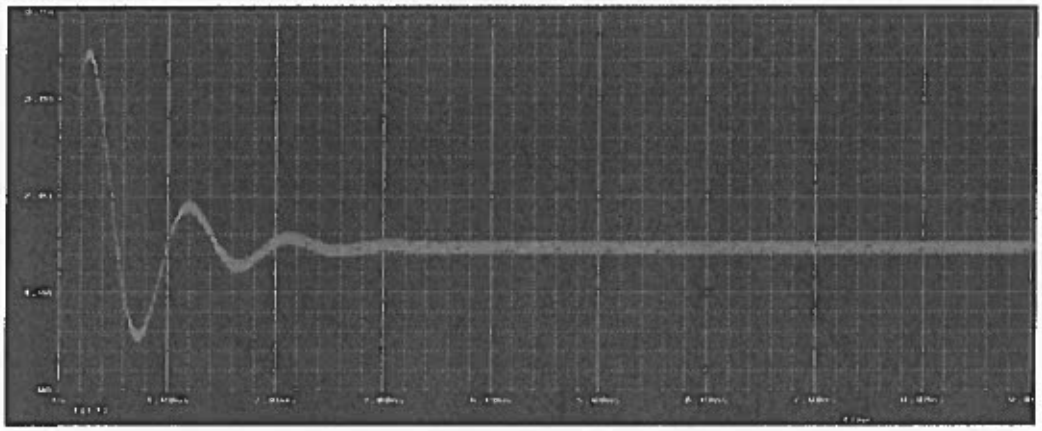


Figure 6

This is the most extreme turn-on case, since the converter must go from zero to maximum load current. We clearly observe a second order damped system behaviour, with exponentially decaying harmonic oscillations.

Table 3

	Max overshoot (%)	Steady-state value	Settling time	Harmonic oscillation frequency/Hz	Rise time
Vout	45.8	48.0V	3.5ms	1053	0.275ms
Inductor I	235%	1.47A	3.5ms	1053	0.08ms

We observe that the current overshoot is much greater than the voltage overshoot and could present a serious problem. Methods of dealing with excess inrush current will have to be found, since such high currents would either damage the components, or require unnecessarily higher rated components in our converter. Settling times and oscillation frequencies are the same for both output voltage and inductor currents.

We wish to test whether this behaviour is expected and also in order to have a better understanding of the system. For this purpose we will derive a dynamic transfer function model for the ideal boost converter using the averaging technique and then apply small-signal perturbations to obtain the small-signal model. The system equations obtained using averaging are as follows:

$$\overline{V_O} = (1-\Delta)\overline{i_L}R_O - RC\frac{d\overline{V_O}}{dt}$$

$$\overline{V_{IN}} = L\frac{d\overline{i_L}}{dt} + (1-\Delta)\overline{V_O}$$

Apply small-signal perturbations →→

$$(\overline{V_O} + \widetilde{V_O}) = (1-\Delta-\tilde{\delta})(\overline{i_L} + \widetilde{i_L})R_O - RC\frac{d}{dt}(\overline{V_O} + \widetilde{V_O})$$

$$(\overline{V_{IN}} + \widetilde{V_{IN}}) = (1-\Delta-\tilde{\delta})(\overline{V_O} + \widetilde{V_O})R_O + L\frac{d}{dt}(\overline{i_L} + \widetilde{i_L})$$

Extract the DC equation

$$\overline{V_O} = \frac{1}{1-\Delta}\overline{V_{IN}}$$

Extract AC equations

$$\widetilde{V_O} = \widetilde{i_L}R_O - \Delta\widetilde{i_L}R_O - \tilde{\delta}\overline{i_L}R - RC\frac{d\widetilde{V_O}}{dt}$$

$$\widetilde{V_{IN}} = L\frac{d\widetilde{i_L}}{dt} + \widetilde{V_O} - \Delta\widetilde{V_O} - \tilde{\delta}\overline{V_O}$$

Apply Laplace Transform

$$\widetilde{V_O} = \widetilde{i_L}R_O(1-\Delta) - \tilde{\delta}\overline{i_L}R - sRC\widetilde{V_O}$$

$$\widetilde{V_{IN}} = sL\widetilde{i_L} + (1-\Delta)\widetilde{V_O} - \tilde{\delta}\overline{V_O}$$

Combine the results into transfer function

$$\widetilde{V_O} = \frac{\frac{\widetilde{V_{IN}}}{1-\Delta} + (\frac{\overline{V_{IN}}}{(1-\Delta)^2} - \frac{sL\overline{V_{IN}}}{R_O(1-\Delta)^4})\tilde{\delta}}{1 + s\frac{L}{R_O(1-\Delta)^2} + s^2\frac{LC}{(1-\Delta)^2}}$$

This transfer function represents a two input system. If we wish to study the nature of the output voltage overshoot when we apply input voltage (which we do), then we must consider that $\tilde{\delta} = 0$, and the transfer function reduces to a traditional second order system. However, note that if we consider the input constant and the duty cycle time-varying (which is what we will do ultimately), then the output is a second order system with a far RHP zero. This RHP zero may affect the stability of the system if we were to apply feedback to the system. For now we set the duty cycle variations to zero. Comparing the obtained transfer function to that of a second order system we may obtain the damping factor expression.

$$\varsigma = \frac{\sqrt{(LC)}}{2R_OC(1-\Delta)}$$

Another useful observation is that the frequencies of the harmonic oscillations are entirely contained by $\frac{1}{2\pi}$ x (imaginary part of the pole). Also:

$$t_{rise} = \frac{1}{w_n}\frac{1}{\sqrt{(1-\varsigma)}}\{\pi - \arctan(\frac{\sqrt{(1-\varsigma^2)}}{\varsigma})\}$$

$$t_{settle} = -\frac{\ln(tolerance\ ratio * \sqrt{(1-\varsigma^2)})}{\varsigma w_n}$$

$$w_n = \frac{(1-\Delta)}{\sqrt{(LC)}}$$

We compute the above values for our above system. Let tolerance ratio be 1%. This yields:

Zeta = 0.237, Wn =6945, trise = 0.298ms, tsettle = 2.80ms, solve denominator for poles →f = 1074Hz

If we look back at table 3, we realise that these results are actually extremely close to the simulated ones. We may now use the better understanding of the system to our advantage.

Change in load

Figure 7 shows the output voltage when we switch from 120 Ohms to 80 Ohms load (50%). Figure 8 shows the output voltage when we switch from 160 Ohms to 80 Ohms (100%).

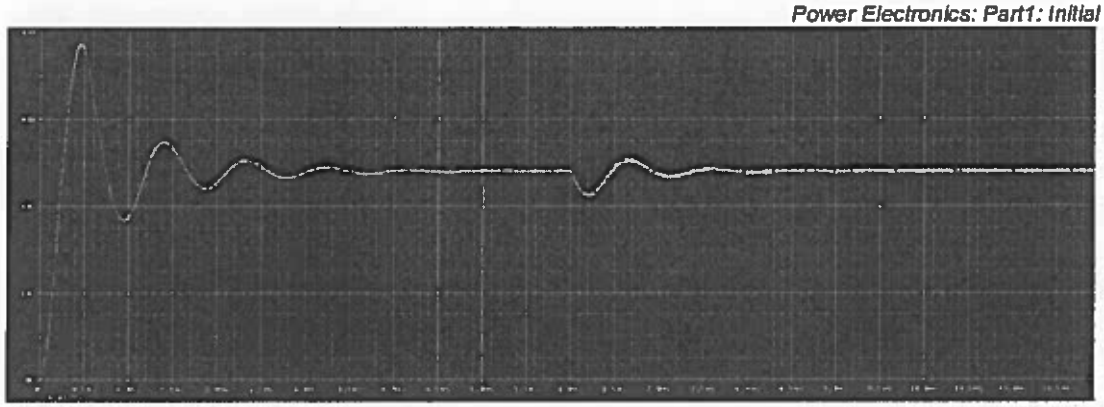


Figure 7

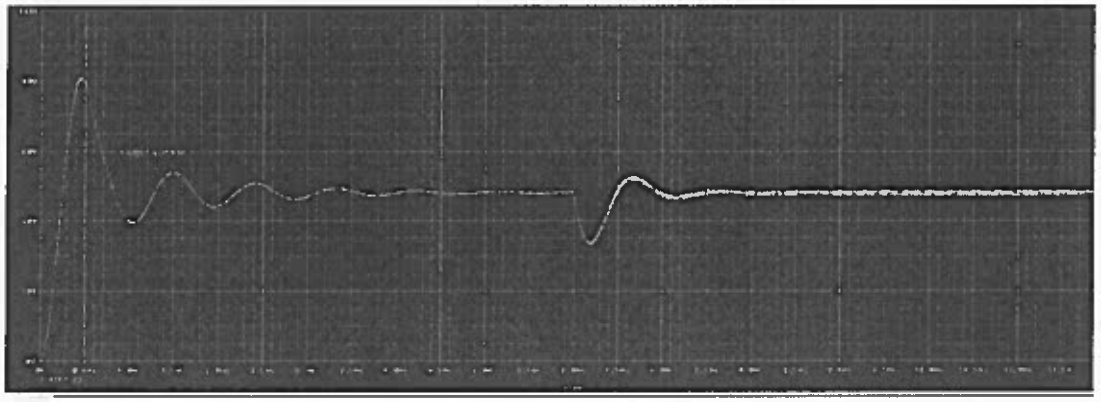


Figure 8

We know from our model analysis that the dc output voltage is a function of the input voltage, the duty cycle and the forward diode voltage drop only. It should ideally not depend on the load and thus on the output current. This is exactly what we observe for both 50% and 100% load switches. After the load switches at 6ms and the oscillations die down the dc output voltage returns to the expected steady state. However, the output voltage ripple increases for lower loads. This is expected, since the output capacitor voltage ripple is proportional to the dc output current.

Low loads

For very low loads the output voltage appears lower than we would have expected. The inductor current ripple becomes larger, along with the inductor current itself. There is a drop in efficiency, since more power is dissipated in the diode during the off-cycle. However, this behaviour of the SMPS should theoretically not be important to us, since the lowest load we may have is 80 Ohms (output current = 0.6A, specification 3) and the converter performs exactly as we would have expected it to.

4 Real Component Selection

Inductor

We have previously argued in section 2 that in a design which contains inefficiencies (a realistic design), the inductor value could be chosen to even be less than the calculated value of 907uH, and produce an acceptable ripple size. It is more advisable, however, to choose a larger inductor than 907uH, both to introduce a safety margin on the potential ripple current, and also because larger inductors will have larger RMS and saturation ratings. These are desirable since we may be confident that the inductor will operate as expected even if large transients are present. However, we are limited by the 12cm^3 core volume requirement. Given our limit on inductor current ratings, the volume requirement effectively places a realistic 1mH threshold on the size of the inductor. The following inductor is considered:

Model name	Store	Inductance	Maximum series resistance	RMS current rating	Saturation current	Core diameter	Core height
Coilcraft PCV-2-105-02L	Farnell	1mH, 10% tolerance	0.37Ohms	2.3A	"high"	18.29mm	21.08mm

The above inductor achieves all of the maximum steady-state requirements. However, inrush current limiting will have to be introduced to ensure that the input dc current never exceeds 2.3A during transient behaviour. The core volume can be calculated as $V = \frac{\pi d^2}{4} h = \frac{\pi \cdot (1.829)^2}{4} \cdot 2.108 = 5.54cm^3$, which is smaller than 12.

Capacitor

The most difficult component to select was the capacitor. Our calculations estimate that the minimum size is 3.8uF in order to achieve 1V output voltage ripple, which is small. However, our model did not take capacitor in-series resistance into account. Thus, even a small hint of ESR would compromise the behaviour. Two solution exist. We could choose a capacitor type with the lowest possible ESR, or choose a much larger aluminium electrolytic capacitor. A larger capacitor would naturally reduce the output ripple due to the capacitance, thus giving more headroom for the ripple due to the ESR.

Before we begin, we must understand that the ESR is a frequency-dependent parameter, as well as temperature-dependent. ESR also has different types of behaviour for different types of capacitor. For electrolytic capacitors, the ESR reduces with frequency, a property which is particularly useful to us. Effectively a large enough electrolytic capacitor would solve our problem. Smaller electrolytic capacitors, for example 10uF, have extremely high ESRs, of the order of 10 Ohms, which is unacceptable for us. Placing several identical capacitors in parallel could also help reduce the ESR at the expense of a higher capacitance and board space.

Choosing naturally low ESR capacitors would provide the optimum solution and reduce the minimum capacitor size. These would have to be either ceramic or film capacitors. However, even multilayer ceramic capacitors usually come in very small sizes, even 3uF is uncommon, and film capacitors have a dissipation factor which increases with frequency. Both of these types are not useful to our applications, so we decide to choose a low-ESR electrolytic capacitor of a larger size than originally expected.

The Vishay 136 RVI is a family of low ESR electrolytic capacitors. The 100 Volt-rated 100uF capacitor is the minimum capacitor from that family that satisfies all our requirements. The ESR at 100Hz can be calculated using δ, the dissipation factor, with the expression $ESR = \frac{tan\delta}{2\pi fC} = \frac{0.07}{2\pi \cdot 100 \cdot 100u} = 1.110hms$. The datasheet provides a typical ESR multiplier plot as a function of frequency. Thus at 100kHz frequency, the ESR should be expected to be

The datasheet also provides a typical ESR multiplier plot, which can be seen in figure 9. Thus we expect our capacitor to have an ESR of approximately 0.2*1.110 = 0.222 Ohms. This value will satisfy our ripple requirement. This capacitor is also large enough to handle the maximum possible RMS current ripple through it (measured at 724mA at max load), since its RMS current ripple rating is 800mA. It is very important that this rating is satisfied, since a higher RMS ripple may damage the component due to increased heating losses in the parasitic resistances during boost operation.(capacitor would heat up more than allowed for safe operation).

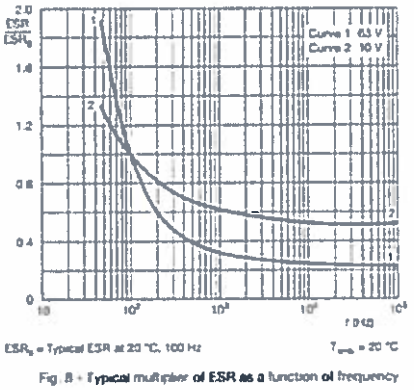


Fig. 8 - Typical multiplier of ESR as a function of frequency

Figure 9

MOSFET

The MOSFET choice must be selected for lower drain-source resistance to minimise power loss, and fast switching time to ensure that it can operate at a frequency of 100kHz. We must also check that all the previously-defined steady-state maximum

ratings have been satisfied, and that the MOSFET would also be able to handle the transient response in the system → must have a safety margin. A good compromise between all these parameters is provided by the FDPF680N10T N-channel MOSFET. It is rated at 100V, 12A with low drain-source resistance of 68mOhms. Switching performance is also listed as "superior" in the datasheet.

Diode

Since the drain-source resistance of the MOSFET is small, and the switching losses are also expected to be small due to its datasheet specifications, the main source of inefficiency within our design will be created by the diode due its non-zero forward voltage drop. In order to minimise the diode loss we must choose a power diode with as small voltage drop as possible, while still making sure that it achieves the current and voltage blocking limits. Hence we decide to use a power Schottky diode, which has been particularly tailored towards improved switching performance. Schottky diodes have a lower on-state voltage drop than semiconductor-semiconductor junction diodes. The STPS8H100FP diode achieves all of these specifications. We expect a voltage drop no higher than approximately 0.4 V.

Gate driver

We should choose a double gate driver IC with a half-bridge configuration, such that both high-side and low-side MOSFETs could be switched. This will allow us to achieve maximum efficiency by converting our design into a synchronous boost converter and minimising diode loss (replace diode with another MOSFET). An all-time popular choice for this application would be the IR2110, which also has pspice simulation support.

5 Simulations

After importing the manufacturer-sourced models for the diode and the MOSFET we may simulate our system again with real component values. Please note that the gate driver will be used in the next section. For now we use the Vpulse source for MOSFET switching. Figure 10 shows the circuit configuration used. Component values have been adjusted accordingly and parasitic estimations added from the datasheets.

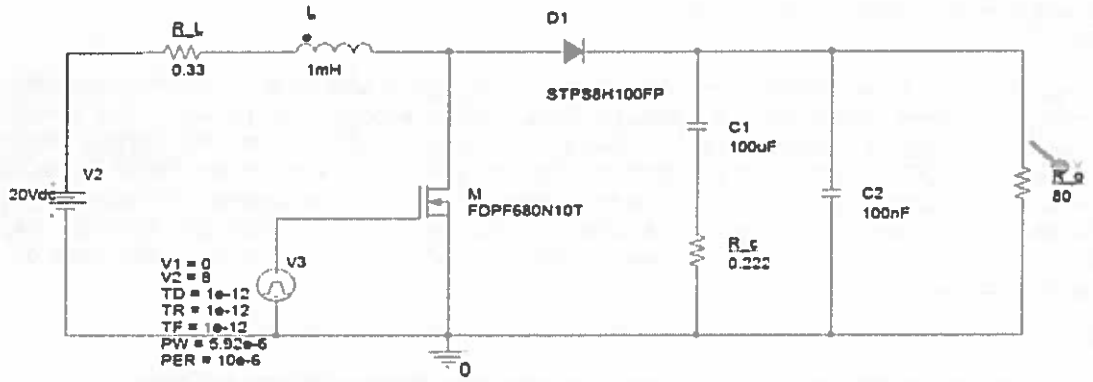


Figure 10

A 100nF ceramic capacitor has also been added in parallel with the 100uF electrolytic. The smaller capacitor is used to remove very frequency oscillations from output voltage which occur as a result of component parasitics. It will also help improve the frequency response of the larger output capacitor. Since C2 is ceramic, its ESR is extremely small (less than a few milliohms) and does not have to be considered. It is also very small compared to C1 and its contribution will only be relevant in much higher frequency bands. We may thus ignore it completely for the 100kHz frequencies that we are operating at. Figure 11 shows the output voltage with C2, Figure 12 shows output voltage without C2. Figure 13 shows the high-frequency oscillations occurring in Figure 12. We immediately see the benefits of adding C2.



Figure 12

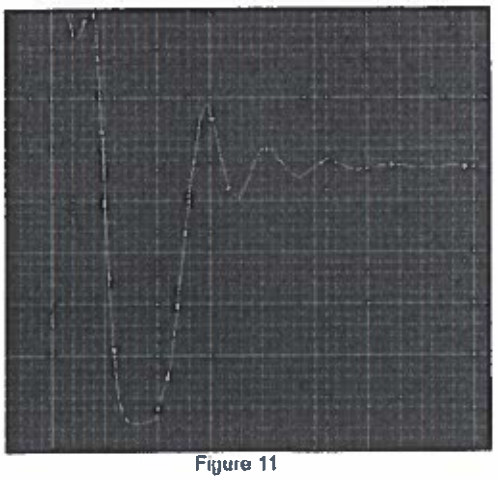


Figure 11

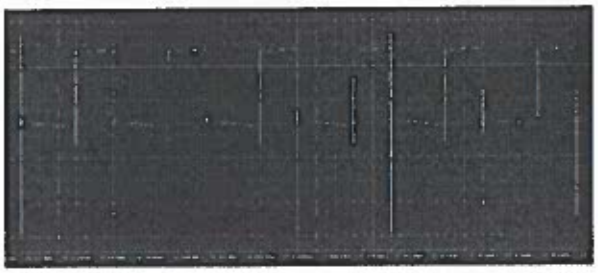


Figure 13

	efficiency
R1=80Ohms→Io=0.6A	η= 0.956
R1=96Ohms→Io=0.5A	η= 0.960
R1=120Ohms→Io=0.4A	η= 0.965
R1=160Ohms→Io=0.3A	η= 0.969
R1=240Ohms→Io=0.2A	η= 0.971
R1=480Ohms→Io=0.1A	η= 0.973

Table 4

We note that, as before, the efficiency of the boost is very high. It also starts to drop for increasing load currents, since more heat is being dissipated in the parasitics, as well as in the diode. This is expected behaviour. The forward voltage drop of the diode is approximately 0.5V, as expected. Average dissipated power in the MOSFET is measured at 170mW at maximum load current, which is very small. Our MOSFET is indeed very efficient, since this is only 0.6% of the output power (negligible losses). This quantity includes both the switching and the drain-source resistance losses. Since both the diode and MOSFET are very efficient, the overall converter efficiency has remained almost the same as in the "ideal" case we had considered before.

Figure 14 shows the output voltage transient at device turn-on for maximum load current.

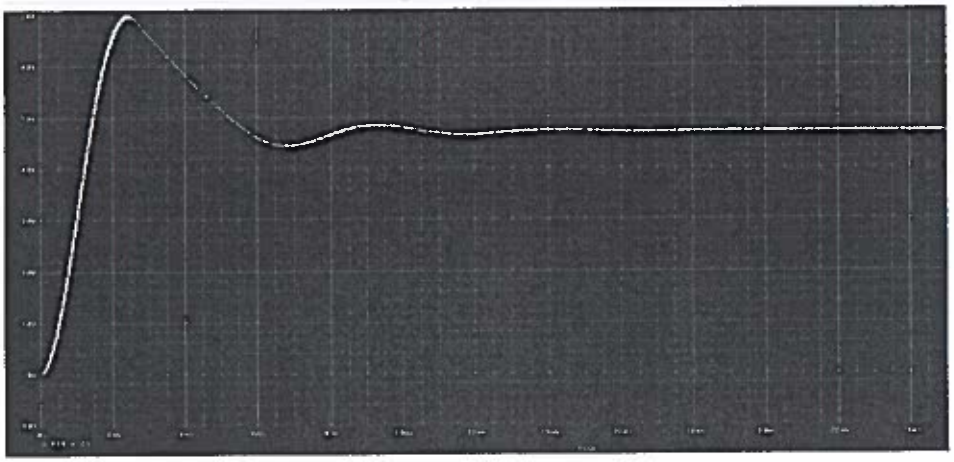


Figure 14

	Max overshoot (%)	Steady-state value	Settling time	Harmonic oscillation frequency/Hz	Rise time
Vout	48.8	47.0V	18ms	196	1.4ms

Table 5

We note that there are differences between this system and the previous. The steady-state value is lower than predicted by our equations (48V) since there has been a drop in efficiency, which as a result gets reflected in a lower output voltage. The maximum overshoot is very slightly higher. But, most importantly, our system is now much slower due to the increase in capacitor value from 3.8uF to 100 uF. Thus rise time and settling times are higher and frequency is lower.

Figure 15 shows the switch from 120 Ohm load to 80 Ohms, and figure 16 shows the switch from 160 Ohm load to 80 Ohm load.

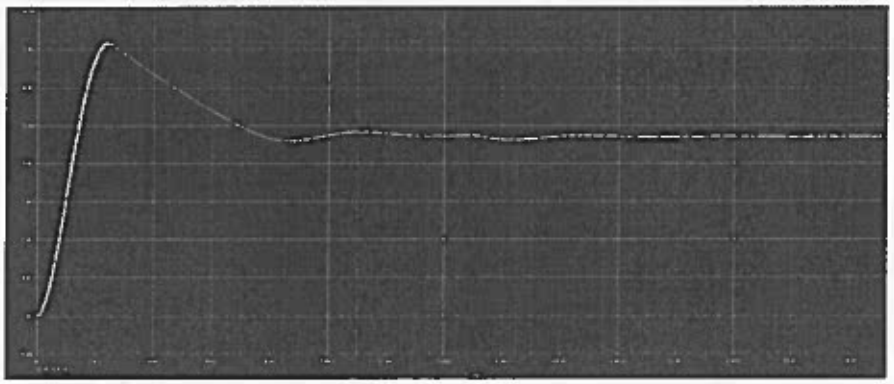


Figure 15

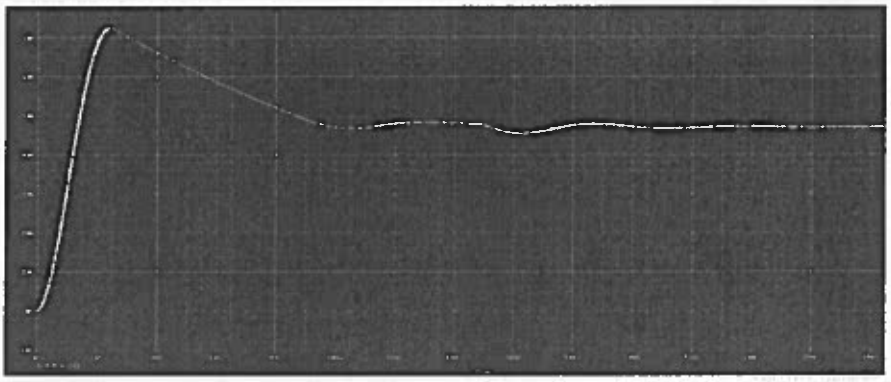


Figure 16

There are several things to note. After the overshoot there is a section of output voltage which looks like exponential decay. This is due to the fact that the capacitor can only discharge as quickly as the effective time constant allows it to, which is in turn determined by the size of the output load. This is exactly why for the higher resistor value in Figure 16 the capacitor takes a lot longer to approach steady-state. We did not see this behaviour previously, since the capacitor was much smaller and it could discharge itself through the resistor at faster rates than the damped harmonic oscillations. It thus appears that our system has a combination of a damped and harmonic responses.

Another very important point to note is that in the real system the steady-state voltage becomes a function of the output current too. Thus when we switch to higher loads, the voltage settles to a lower dc level than we would have originally expected it to. This is to do with lower efficiency at higher currents, which is reflected in lower steady-state output voltages, and hence lower output power to satisfy the power conservation law. This is why the notion of feedback is particularly applicable in reality!

For low loads the efficiency drops dramatically, steady-state output voltages become lower than expected, and very large-amplitude ripple currents appear at the output of the converter.

6 Synchronous rectification

Our design will perform less efficiently in reality than in the simulation. Thus it is best to try and minimise the component power dissipation losses. We have previously determined in our analysis that the largest share of inefficiency is due to the forward drop of the diode. This problem could be avoided if the diode was replaced by another MOSFET, which would be synchronously switched such that the circuit behaves the same way. The benefit arises from the fact that the voltage drop across the MOSFET we are using is considerably less than the forward voltage drop of the diode that we are using, which will result in lower power loss. Since the two MOSFETS will have to be synchronised with each other (only one MOSFET should be on at any particular time instance) the best solution would be to use the half-bridge rectifier IC that we had picked earlier, i.e. the IR2110. The notion of a half-bridge is directly applicable to our circuit, since the syncFET will behave exactly like a high-side device, and the other MOSFET will behave as a low-side device. In order to swtich the high-side device, the IC must provide a gate voltage for the syncFET which is referenced to the syncFET's source, and not ground (remember that the state of the MOSFET is controlled by the gate-source voltage, and not gate-ground). This is because the source is floating and is at the same potential as the drain of the switching MOSFET. A typical application schematic for the IR2110 half bridge implementation is provided in its datasheet.

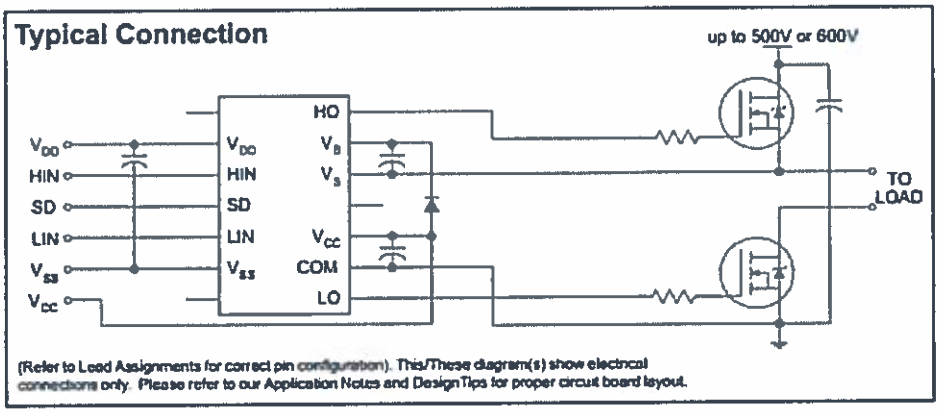


Figure 18

The purpose of the Vb-Vs "bootstrapping" capacitor is to contain the contain the voltage which will be added to the source of the syncFET to produce the gate voltage for the syncFET. Mapping the circuit on figure 18 onto our boost converter application results in the circuit shown on Figure 17.

There are several points which have to be noted when simulating this model. Possibly due to bugs in the model for the IR2110, the circuit will not simulate and give a convergence error unless:

- The diode between Vcc and Vb is removed.

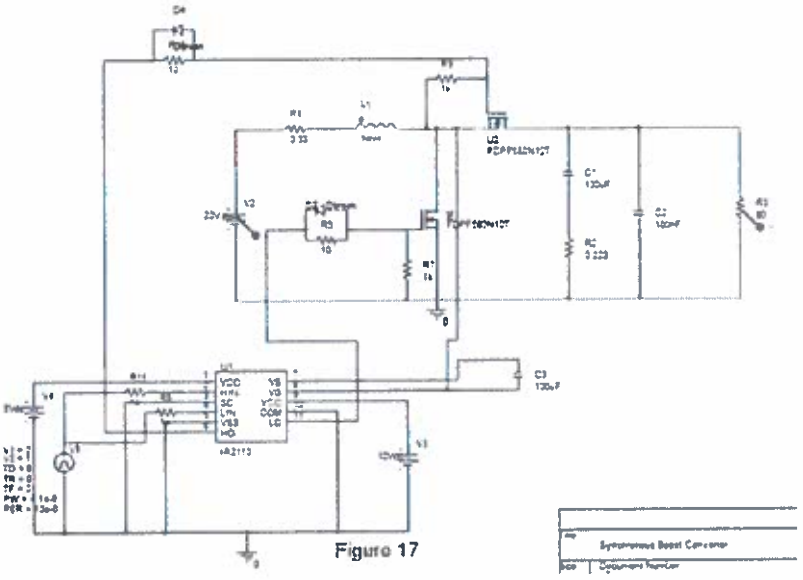


Figure 17

- All the initial conditions of components with memory are defined
- The option "Skip the initial bias point calculation" is ticked in the simulation editor.

Make sure to select a maximum step size of at least 5us for correct simulation results. The simulation takes a long time to simulate due to the number of components in the design. We test our circuit to see if the behaviour is still the same. Figure 19 shows the output voltage of the system.

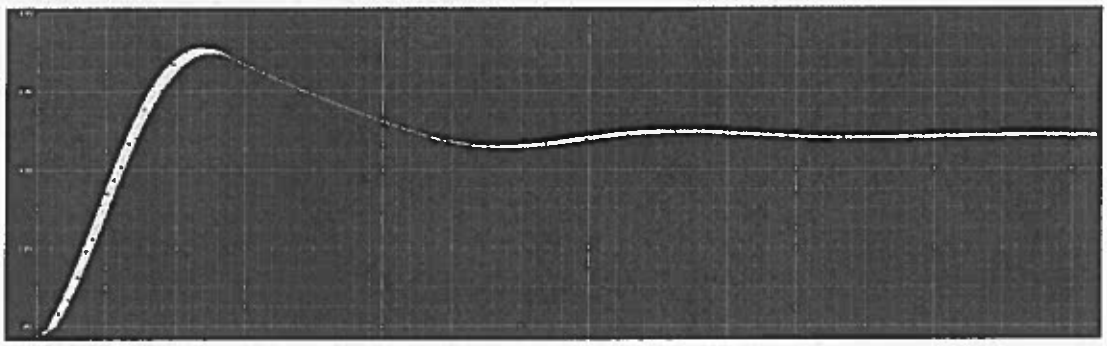


Figure 19

We see that the circuit behaves as we would have normally expected it to. However, due to the nature of the MOSFET the inductor current in discontinuous mode is allowed to go slightly negative (-22mA). However, this in to an issue which will particularly affect our system, since we will be operating in continuous mode for almost all of the time (we must do if we are to achieve the specification!). Figure 20 shows the discontinuous inductor current. We note the similarity of Figure 19 to Figure 14.

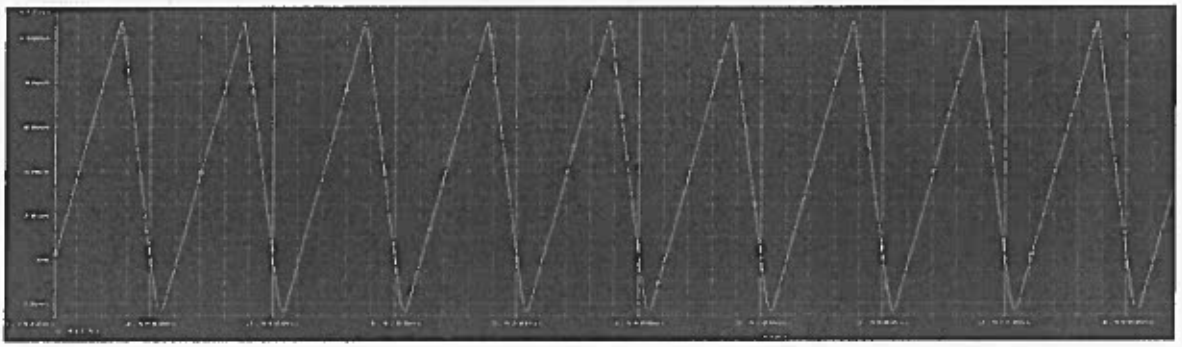


Figure 20

We should now estimate whether we had made any significant improvements to system performance. We estimate the efficiency for the maximum output load case, i.e. an output current of 0.6A. The efficiency of the power supply is again high at 95.3%. However, this time we are using substantially more components and are also using a real gate drive IC to switch the MOSFETs on and off, as opposed to an ideal S component. Even though the simulations show similar efficiencies, we expect this circuit to operate more efficiently than the version with the diode and will attempt to use the synchronous principle when building the converter.

7 Closed-loop system

We analyse our open-loop system in Matlab using the averaged state-space model used in the lecture notes. The model does not include the forward voltage drop of the diode. But, the synchronous boost we are designing the feedback for has only a small voltage across the syncFET, so this assumption is reasonable. Figure 21 shows the frequency response of our open-loop system. Figure 22 shows the pole-zero plot for the system. The double pole is has a real part of -267.2 and the RHP zero at around 10000. The far LHP zero is not shown on the plot and is not essential to the design of the feedback controller.

A guideline for converters with RHP zero is to design at the lowest input voltage and the maximum load. This causes the lowest value of RHP zero, and the lowest value of resonant frequency. When this happens, phase

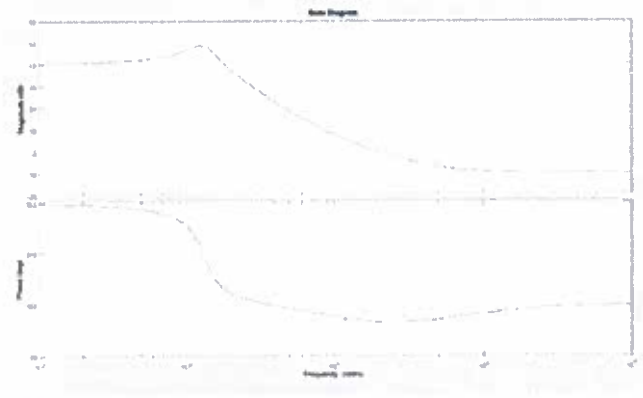


Figure 22

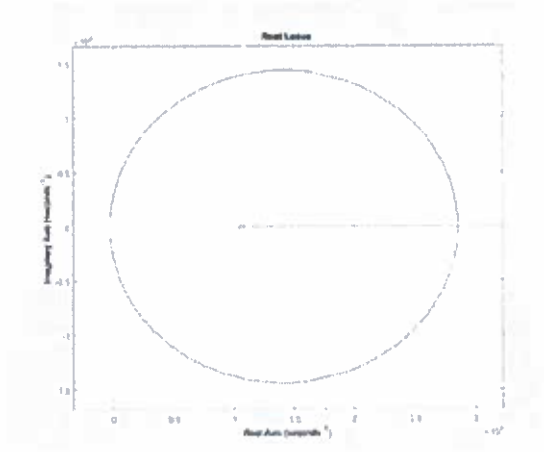


Figure 22

margin is at minimum, hence higher input voltages would improve stability. Making sure that our system is stable for the worst possible case would ensure that it is stable for most cases. For this reason we have selected the large signal parameters as Vin = 18V, Rout = 80Ohms, Delta = 0.640(so that output voltage is as close to 48V dc as possible).

For the control loop design we attempt to use a PI controller, as outlined in the lecture notes. Figure 22 shows the root locus for the uncompensated system.

We note that the root locus provides us with little options, since almost no values of k provide us with a stable system with an acceptable value of the damping ratio. We add a PI controller and introduce a zero in the LHP to shift the root locus leftwards. However, no matter the location of the zero, this does not lead to any significant improvements in the situation.

We have not yet been able to find a solution to this issue. We must thoroughly reconsider the model to improve its stability, as well as consider using different types of compensators.

Power Electronics Coursework

GRADEMARK REPORT

FINAL GRADE

/100

GENERAL COMMENTS

Instructor

PAGE 1

Comment 1

Total grade: B

Converter design: B
Open Loop: A
Component Choices: A
Real Components: A
Synchronous Rectification: B
Closed Loop: E
Additional Marks: C

PAGE 2

PAGE 3

Comment 2

It's a dc output...

Comment 3

?

Comment 4

Not necessarily

Comment 5

Again. Not necessarily

Comment 6

?

PAGE 4

Comment 7

Trivial

Comment 8

inductance

Comment 9
inductance

Comment 10
You should be much clearer in what you write

PAGE 5

Comment 11
In the power range that you are working, it is not true.

Comment 12
You are not here to guess

Comment 13
All of this is said without any proven research, which makes it not valid.

Comment 14
Why?

Comment 15
If you feed a resistor with a voltage that has a ripple, the current in the resistor will have a ripple... V = RI

Comment 16
Why? ref?

Comment 17
This is not true

PAGE 6

Comment 18
?

Comment 19
Not true

Comment 20
?

PAGE 7

PAGE 8

PAGE 9

PAGE 10

Comment 21
You should do a better analysis of this.

Comment 22
low resistance or low power?

Comment 23
It is to understand the concept of overloading the SMPS. The reason why the efficiency drops should be explained if this is a case you are analysing in the report

PAGE 11

Comment 24
?

Comment 25
This will be a significant source of losses. If you designed for 200kHz, your inductance requirements would be lower, and you could find more efficient components.

Comment 26
These type of comments should not be in the report

Comment 27
not true http://uk.farnell.com/webapp/wcs/stores/servlet/Search?catalogId=15001&langId=44&storeId=10151&categoryId=700000005423&st=x7r%20capacit

PAGE 12

Comment 28
C_out might be a little bit to high

Comment 29
Good. Why didn't you go back and fix the statement of Vf = 1V

Comment 30
ok!

Comment 31
Why? This contradicts previous statements of this report

PAGE 13

Comment 32
You said no more than 0.4V...

Comment 33
Nope!

PAGE 14

PAGE 15

Comment 34
good

PAGE 16

PAGE 17